Class 10 Mathematics — Exercise 2.2

Question 1

Reduce the following equations to quadratic form using suitable substitution.

Standard quadratic form:

$$a x^2 + bx + c = 0$$

(i)

$$a x^4 + b x^2 + c = 0 \quad \ldots(1)$$

Let

$$x^2 = y$$
$$\left(x^2\right)^2 = y^2$$
$$x^4 = y^2$$

Put in equation (1):

$$a y^2 + by + c = 0 \quad \textbf{(Quadratic form)}$$

(ii)

$$9 x^6 - 3 x^3 + 7 = 0 \quad \ldots(1)$$

Let

$$x^3 = y$$
$$\left(x^3\right)^2 = y^2$$
$$x^6 = y^2$$

Put in equation (1):

$$9 y^2 - 3 y + 7 = 0 \quad \textbf{(Quadratic form)}$$

(iii)

$$3x + \frac{4}{6x - 2} = -1$$

Take 2 common from the denominator:

$$3x + \frac{4}{2(3x - 1)} = -1$$
$$3x + \frac{2}{3x - 1} = -1$$

Put

$$3x = y$$
$$y + \frac{2}{y - 1} = -1$$

Multiply both sides by (y − 1):

$$y(y-1)+\frac{2}{y-1}\times(y-1)=-1(y-1)$$

$$y^2-y+2=-y+1$$

$$y^2-y+y+2-1=0$$

$$y^2+1=0 \quad \textbf{(Quadratic form)}$$

(iv)

$$(x+1)^2+\frac{3}{(x+1)^2}=4$$

Let

$$(x+1)^2=z$$

$$z+\frac{3}{z}=4$$

Multiply both sides by z:

$$z\times z+\frac{3}{z}\times z=4z$$

$$z^2+3=4z$$

$$z^2-4z+3=0 \quad \textbf{(Quadratic form)}$$

(v)

$$3\left(x^2+\frac{1}{x^2}\right)+8\left(x+\frac{1}{x}\right)+11=0$$

Let

$$x+\frac{1}{x}=y$$

Square both sides:

$$\left(x+\frac{1}{x}\right)^2=y^2$$

$$x^2+\left(\frac{1}{x}\right)^2+2(x)\left(\frac{1}{x}\right)=y^2$$

$$x^2+\frac{1}{x^2}+2=y^2$$

$$x^2+\frac{1}{x^2}=y^2-2$$

Put in equation (1):

$$3(y^2-2)+8(y)+11=0$$

$$3y^2-6+8y+11=0$$

$$3y^2+8y+5=0 \quad \textbf{(Quadratic form)}$$

(vi)

$$3x^4+7x^3+5x^2-7x+3=0$$

Divide each term by x^2:

$$\frac{3x^4}{x^2}+\frac{7x^3}{x^2}+\frac{5x^2}{x^2}-\frac{7x}{x^2}+\frac{3}{x^2}=\frac{0}{x^2}$$

$$3x^2+7x+5-\frac{7}{x}+\frac{3}{x^2}=0$$

Rearrange the terms:

$$3x^2+\frac{3}{x^2}+7x-\frac{7}{x}+5=0$$

$$3\left(x^2+\frac{1}{x^2}\right)+7\left(x-\frac{1}{x}\right)+5=0 \quad ...(1)$$

Let

$$x-\frac{1}{x}=z$$

Square both sides:

$$\left(x-\frac{1}{x}\right)^2=z^2$$

$$x^2+\left(\frac{1}{x}\right)^2-2(x)\left(\frac{1}{x}\right)=z^2$$

$$x^2+\frac{1}{x^2}-2=z^2$$

$$x^2+\frac{1}{x^2}=z^2+2$$

Put in equation (1):

$$3(z^2+2)+7(z)+5=0$$

$$3z^2+6+7z+5=0$$

$$\boxed{3z^2+7z+11=0 \quad \textbf{(Quadratic form)}}$$

(vii)

$$ak^{2x}+bk^x+c=0 \quad ...(1)$$

Let

$$k^x=z$$

$$(k^x)^2=z^2$$

$$k^{2x}=z^2$$

Put in equation (1):

$$\boxed{az^2+bz+c=0 \quad \textbf{(Quadratic form)}}$$

(viii)

$$8\times 4^{x}-7\times 2^{x}-1=0 \quad ...(1)$$

1st Method

Let

$$2^{x}=y$$
$$\left(2^{x}\right)^{2}=y^{2}$$
$$2^{2x}=y^{2}$$
$$4^{x}=y^{2}$$

Put in equation (1):

$$8y^{2}-7y-1=0 \quad \textbf{(Quadratic form)}$$

2nd Method

Since $4 = 2^2$, write:

$$4^{x}=\left(2^{2}\right)^{x}=2^{2x}$$
$$8\times 2^{2x}-7\times 2^{x}-1=0 \quad ...(1)$$

Let

$$2^{x}=y$$
$$\left(2^{x}\right)^{2}=y^{2}$$
$$2^{2x}=y^{2}$$

Put in equation (1):

$$8y^{2}-7y-1=0 \quad \textbf{(Quadratic form)}$$

(ix)

$$(x-1)(x-2)(x+3)(x-6)=6$$

Pair the factors so that the constant terms add up to the same number: $(-1) + (-2) = -3$ and $(+3) + (-6) = -3$

$$\left[(x-1)(x-2)\right]\left[(x+3)(x-6)\right]=6$$
$$\left[x(x-2)-1(x-2)\right]\left[x(x-6)+3(x-6)\right]=6$$
$$\left(x^{2}-2x-x+2\right)\left(x^{2}-6x+3x-18\right)=6$$
$$\left(x^{2}-3x+2\right)\left(x^{2}-3x-18\right)=6$$

Let

$$x^{2}-3x=y$$
$$(y+2)(y-18)=6$$
$$y(y-18)+2(y-18)-6=0$$
$$y^{2}-18y+2y-36-6=0$$

$$y^{2}-16y-42=0 \quad \textbf{(Quadratic form)}$$

(x)

$$(2x-1)(2x-7)(x-3)(x-1)=8$$

1st Method

Take 2 common from (2x − 1) and (2x − 7):

$$2\left(x-\frac{1}{2}\right)2\left(x-\frac{7}{2}\right)(x-3)(x-1)=8$$

$$4\left(x-\frac{1}{2}\right)\left(x-\frac{7}{2}\right)(x-3)(x-1)=8$$

Pairing check: −1/2 − 7/2 = −8/2 = −4 and −3 − 1 = −4

$$4\left[\left(x-\frac{1}{2}\right)\left(x-\frac{7}{2}\right)\right][(x-3)(x-1)]=8$$

$$4\left[x\left(x-\frac{7}{2}\right)-\frac{1}{2}\left(x-\frac{7}{2}\right)\right][x(x-1)-3(x-1)]=8$$

$$4\left[x^2-\frac{7}{2}x-\frac{1}{2}x+\frac{7}{4}\right][x^2-x-3x+3]=8$$

$$4\left[x^2-4x+\frac{7}{4}\right][x^2-4x+3]=8$$

Let

$$x^2-4x=z$$

$$4\left(z+\frac{7}{4}\right)(z+3)=8$$

$$(4z+7)(z+3)=8$$

$$4z(z+3)+7(z+3)=8$$

$$4z^2+12z+7z+21-8=0$$

$$\boxed{4z^2+19z+13=0 \quad \textbf{(Quadratic form)}}$$

2nd Method

$$[(2x-1)(2x-7)][(x-3)(x-1)]=8$$

$$[2x(2x-7)-1(2x-7)][x(x-1)-3(x-1)]=8$$

$$(4x^2-14x-2x+7)(x^2-x-3x+3)=8$$

$$(4x^2-16x+7)(x^2-4x+3)=8$$

$$(4(x^2-4x)+7)(x^2-4x+3)=8$$

Let

$$x^2-4x=z$$

$$(4z+7)(z+3)=8$$

$$4z(z+3)+7(z+3)=8$$

$$\boxed{4z^2+19z+13=0 \quad \textbf{(Quadratic form)}}$$

Question 2

Solve the following equations by reducing them to quadratic form.

(i)

$$x^4 - 20x^2 + 64 = 0 \quad ...(1)$$

Let

$$x^2 = y$$
$$(x^2)^2 = y^2$$
$$x^4 = y^2$$

Put in equation (1):

$$y^2 - 20y + 64 = 0$$

Factorise (split middle term): $(-16) \times (-4) = 64$ and $(-16) + (-4) = -20$

$$y^2 - 16y - 4y + 64 = 0$$
$$y(y - 16) - 4(y - 16) = 0$$
$$(y - 4)(y - 16) = 0$$
$$y - 4 = 0$$
$$y = 4$$
$$x^2 = 4$$
$$\sqrt{x^2} = \sqrt{4}$$
$$x = \pm 2$$

or

$$y - 16 = 0$$
$$y = 16$$
$$x^2 = 16$$
$$\sqrt{x^2} = \sqrt{16}$$
$$x = \pm 4$$

Solution Set: $\{\pm 2, \pm 4\}$

(ii)

$$x^4 + 16x^2 - 225 = 0 \quad ...(1)$$

Let

$$x^2 = y$$
$$(x^2)^2 = y^2$$
$$x^4 = y^2$$

Put in equation (1):

$$y^2 + 16y - 225 = 0$$

Factorise (split the middle term): $25 \times (-9) = -225$ and $25 + (-9) = 16$

$$y^2 + 25y - 9y - 225 = 0$$

$$y(y+25)-9(y+25)=0$$
$$(y-9)(y+25)=0$$

Either

$$y-9=0$$
$$y=9$$
$$x^2=9$$
$$\sqrt{x^2}=\sqrt{9}$$
$$x=\pm 3$$

or

$$y+25=0$$
$$y=-25$$
$$x^2=-25$$
$$\sqrt{x^2}=\sqrt{-25}$$
$$x=\sqrt{25\times(-1)}$$
$$x=\sqrt{25}\ \sqrt{-1}$$
$$x=\pm 5i \quad \textit{(where } i=\sqrt{-1}\textit{)}$$

Solution Set: $\{\pm 3,\ \pm 5i\}$

(iii)

$$x^{\frac{2}{5}}+5x^{\frac{1}{5}}+6=0 \quad \textit{...(1)}$$

Let

$$x^{\frac{1}{5}}=y$$
$$(x^{\frac{1}{5}})^2=y^2$$
$$x^{\frac{2}{5}}=y^2$$

Put in equation (1):

$$y^2+5y+6=0$$
$$y^2+3y+2y+6=0$$
$$y(y+3)+2(y+3)=0$$
$$(y+2)(y+3)=0$$

Either

$$y+2=0$$
$$y=-2$$
$$x^{\frac{1}{5}}=-2$$

Take power 5 on both sides:

$$(x^{\frac{1}{5}})^5=(-2)^5$$

$$x=-32$$

or

$$y+3=0$$
$$y=-3$$
$$x^{\frac{1}{5}}=-3$$
$$\left(x^{\frac{1}{5}}\right)^5=(-3)^5$$
$$x=-243$$

Solution Set: $\{-32,\ -243\}$

(iv)

$$3x^2+\frac{4}{x^2}=7 \quad ...(1)$$

Let

$$x^2=y$$

Put in equation (1):

$$3y+\frac{4}{y}=7$$
$$3y+\frac{4}{y}-7=0$$

Multiply both sides by y:

$$3y\times y+\frac{4}{y}\times y-7y=0$$
$$3y^2+4-7y=0$$
$$3y^2-7y+4=0$$
$$3y^2-3y-4y+4=0$$
$$3y(y-1)-4(y-1)=0$$
$$(3y-4)(y-1)=0$$

Either

$$3y-4=0$$
$$y=\frac{4}{3}$$
$$x^2=\frac{4}{3}$$
$$\sqrt{x^2}=\sqrt{\frac{4}{3}}$$
$$x=\pm\frac{2}{\sqrt{3}}$$

or

$$y-1=0$$

$$y=1$$
$$x^2=1$$
$$\sqrt{x^2}=\sqrt{1}$$
$$x=\pm 1$$

Solution Set: $\{\pm 1,\ \pm\frac{2}{\sqrt{3}}\}$

(v)

$$5(x+1)+\frac{3}{x+1}=8 \quad ...(1)$$

Let

$$x+1=y$$

Put in equation (1):

$$5y+\frac{3}{y}=8$$
$$5y+\frac{3}{y}-8=0$$

Multiply both sides by y:

$$5y^2+3-8y=0$$
$$5y^2-8y+3=0$$
$$5y^2-5y-3y+3=0$$
$$5y(y-1)-3(y-1)=0$$
$$(y-1)(5y-3)=0$$

Either

$$y-1=0$$
$$y=1$$
$$x+1=1$$
$$x=0$$

or

$$5y-3=0$$
$$y=\frac{3}{5}$$
$$x+1=\frac{3}{5}$$
$$x=\frac{3}{5}-1$$
$$x=-\frac{2}{5}$$

Solution Set: $\{0, -\frac{2}{5}\}$

(vi)

$$5x^2+\frac{36}{5x^2+4}=16 \quad ...(1)$$

Let

$$5x^2=y$$

Put in equation (1):

$$y+\frac{36}{y+4}=16$$

Multiply both sides by (y + 4):

$$y(y+4)+36=16(y+4)$$
$$y^2+4y+36=16y+64$$
$$y^2+4y-16y+36-64=0$$
$$y^2-12y-28=0$$
$$y^2+2y-14y-28=0$$
$$y(y+2)-14(y+2)=0$$
$$(y+2)(y-14)=0$$

Either

$$y+2=0$$
$$y=-2$$
$$5x^2=-2$$
$$x^2=-\frac{2}{5}$$
$$x=\pm\sqrt{-\frac{2}{5}}$$

or

$$y-14=0$$
$$y=14$$
$$5x^2=14$$
$$x^2=\frac{14}{5}$$
$$x=\pm\sqrt{\frac{14}{5}}$$

Solution Set: $\{\pm\sqrt{\frac{14}{5}}, \pm\sqrt{-\frac{2}{5}}\}$

(vii)

$$2x^4 - x^3 - 6x^2 - x + 2 = 0$$

Divide each term by x^2:

$$2x^2 - x - 6 - \frac{1}{x} + \frac{2}{x^2} = 0$$

Rearrange the terms:

$$2x^2 + \frac{2}{x^2} - x - \frac{1}{x} - 6 = 0$$

$$2\left(x^2 + \frac{1}{x^2}\right) - 1\left(x + \frac{1}{x}\right) - 6 = 0 \quad \textit{...(1)}$$

Let

$$x + \frac{1}{x} = y$$

Square both sides:

$$\left(x + \frac{1}{x}\right)^2 = y^2$$

$$x^2 + \frac{1}{x^2} + 2(x)\left(\frac{1}{x}\right) = y^2$$

$$x^2 + \frac{1}{x^2} + 2 = y^2$$

$$x^2 + \frac{1}{x^2} = y^2 - 2$$

Put in equation (1):

$$2(y^2 - 2) - 1(y) - 6 = 0$$
$$2y^2 - 4 - y - 6 = 0$$
$$2y^2 - y - 10 = 0$$
$$2y^2 + 4y - 5y - 10 = 0$$
$$2y(y + 2) - 5(y + 2) = 0$$
$$(y + 2)(2y - 5) = 0$$

When y + 2 = 0, i.e. y = −2:

$$x + \frac{1}{x} = -2$$

Multiply both sides by x:

$$x \times x + \frac{1}{x} \times x = -2x$$

$$x^2 + 1 + 2x = 0$$
$$x^2 + 2x + 1 = 0$$
$$x^2 + x + x + 1 = 0$$
$$x(x + 1) + 1(x + 1) = 0$$
$$(x + 1)(x + 1) = 0$$
$$x = -1$$

When 2y − 5 = 0, i.e. y = 5/2:

$$x+\frac{1}{x}=\frac{5}{2}$$

Multiply both sides by 2x:

$$2x\times x+2x\times\frac{1}{x}=\frac{5}{2}\times 2x$$

$$2x^2+2=5x$$

$$2x^2-5x+2=0$$

$$2x^2-4x-x+2=0$$

$$2x(x-2)-1(x-2)=0$$

$$(x-2)(2x-1)=0$$

$$x-2=0 \implies x=2$$

$$2x-1=0 \implies x=\frac{1}{2}$$

Solution Set: $\{-1,\ \frac{1}{2},\ 2\}$

(viii)

$$12x^4+11x^3-146x^2+11x+12=0$$

Divide each term by x^2:

$$12x^2+11x-146+\frac{11}{x}+\frac{12}{x^2}=0$$

Rearrange the terms:

$$12x^2+\frac{12}{x^2}+11x+\frac{11}{x}-146=0$$

$$12\left(x^2+\frac{1}{x^2}\right)+11\left(x+\frac{1}{x}\right)-146=0 \quad ...(1)$$

Let

$$x+\frac{1}{x}=y$$

Square both sides (same as part vii):

$$x^2+\frac{1}{x^2}=y^2-2$$

Put in equation (1):

$$12(y^2-2)+11y-146=0$$

$$12y^2-24+11y-146=0$$

$$12y^2+11y-170=0$$

Split the middle term: $12\times(-170)=-2040=(-40)\times 51$ and $-40+51=11$

$$12y^2-40y+51y-170=0$$

$$4y(3y-10)+17(3y-10)=0$$

$$(3y-10)(4y+17)=0$$

When 3y − 10 = 0:

$$3y=10$$
$$y=\frac{10}{3}$$
$$x+\frac{1}{x}=\frac{10}{3}$$

Multiply both sides by 3x:

$$3x\times x+\frac{1}{x}\times 3x=\frac{10}{3}\times 3x$$
$$3x^2+3=10x$$
$$3x^2-10x+3=0$$
$$3x^2-9x-x+3=0$$
$$3x(x-3)-1(x-3)=0$$
$$(x-3)(3x-1)=0$$
$$x-3=0 \implies x=3$$
$$3x-1=0 \implies x=\frac{1}{3}$$

When 4y + 17 = 0:

$$4y=-17$$
$$y=-\frac{17}{4}$$
$$x+\frac{1}{x}=-\frac{17}{4}$$

Multiply both sides by 4x:

$$4x\times x+\frac{1}{x}\times 4x=-\frac{17}{4}\times 4x$$
$$4x^2+4=-17x$$
$$4x^2+17x+4=0$$
$$4x^2+16x+x+4=0$$
$$4x(x+4)+1(x+4)=0$$
$$(x+4)(4x+1)=0$$
$$x+4=0 \implies x=-4$$
$$4x+1=0 \implies x=-\frac{1}{4}$$

Solution Set: $\{3,\ -4,\ \frac{1}{3},\ -\frac{1}{4}\}$

(ix)

$$4\left(x^2+\frac{1}{x^2}\right)-\left(x-\frac{1}{x}\right)-11=0 \quad \textit{...(1)}$$

Let

$$x - \frac{1}{x} = y$$

Square both sides:

$$\left(x - \frac{1}{x}\right)^2 = y^2$$

$$x^2 + \frac{1}{x^2} - 2(x)\left(\frac{1}{x}\right) = y^2$$

$$x^2 + \frac{1}{x^2} = y^2 + 2$$

Put in equation (1):

$$4(y^2 + 2) - y - 11 = 0$$

$$4y^2 + 8 - y - 11 = 0$$

$$4y^2 - y - 3 = 0$$

$$4y^2 - 4y + 3y - 3 = 0$$

$$4y(y - 1) + 3(y - 1) = 0$$

$$(y - 1)(4y + 3) = 0$$

When y − 1 = 0, i.e. y = 1:

$$x - \frac{1}{x} = 1$$

Multiply both sides by x:

$$x \times x - \frac{1}{x} \times x = 1 \times x$$

$$x^2 - 1 = x$$

$$x^2 - x - 1 = 0$$

Here a = 1, b = −1, c = −1. Using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{1 \pm \sqrt{1 + 4}}{2}$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

When 4y + 3 = 0, i.e. y = −3/4:

$$4y = -3$$

$$y = -\frac{3}{4}$$

$$x - \frac{1}{x} = -\frac{3}{4}$$

Multiply both sides by 4x:

$$4x \times x - \frac{1}{x} \times 4x = -\frac{3}{4} \times 4x$$

$$4x^2 - 4 = -3x$$

$$4x^2 + 3x - 4 = 0$$

Here a = 4, b = 3, c = −4:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3 \pm \sqrt{(3)^2 - 4(4)(-4)}}{2(4)}$$

$$= \frac{-3 \pm \sqrt{9 + 64}}{8}$$

$$x = \frac{-3 \pm \sqrt{73}}{8}$$

Solution Set: $\left\{\frac{1 \pm \sqrt{5}}{2}, \frac{-3 \pm \sqrt{73}}{8}\right\}$

(x)

$$2^{2x} - 34 \times 2^x + 64 = 0 \quad \text{...(1)}$$

Let

$$2^x = y$$

$$(2^x)^2 = y^2$$

$$2^{2x} = y^2$$

Put in equation (1):

$$y^2 - 34y + 64 = 0$$

$$y^2 - 2y - 32y + 64 = 0$$

$$y(y-2) - 32(y-2) = 0$$

$$(y-2)(y-32) = 0$$

Either

$$y - 2 = 0$$

$$y = 2$$

$$2^x = 2^1$$

$$x = 1$$

or

$$y - 32 = 0$$

$$y = 32$$

$$2^x = 32$$

$$2^x = 2^5$$

$$x = 5$$

Solution Set: $\{1, 5\}$

(xi)

$$3^{2x} - 12 \times 3^{x} + 27 = 0 \quad ...(1)$$

Let

$$3^{x} = y$$
$$(3^{x})^{2} = y^{2}$$
$$3^{2x} = y^{2}$$

Put in equation (1):

$$y^{2} - 12y + 27 = 0$$
$$y^{2} - 3y - 9y + 27 = 0$$
$$y(y-3) - 9(y-3) = 0$$
$$(y-3)(y-9) = 0$$

Either

$$y - 3 = 0$$
$$y = 3$$
$$3^{x} = 3^{1}$$
$$x = 1$$

or

$$y - 9 = 0$$
$$y = 9$$
$$3^{x} = 9$$
$$3^{x} = 3^{2}$$
$$x = 2$$

Solution Set: $\{1, 2\}$

(xii)

$$5^{2x} - 150 \times 5^{x} + 3125 = 0 \quad ...(1)$$

Let

$$5^{x} = y$$
$$(5^{x})^{2} = y^{2}$$
$$5^{2x} = y^{2}$$

Put in equation (1):

$$y^{2} - 150y + 3125 = 0$$
$$y^{2} - 25y - 125y + 3125 = 0$$
$$y(y-25) - 125(y-25) = 0$$
$$(y-25)(y-125) = 0$$

Either

$$y-25=0$$
$$y=25$$
$$5^x=25$$
$$5^x=5^2$$
$$x=2$$

or

$$y-125=0$$
$$y=125$$
$$5^x=125$$
$$5^x=5^3$$
$$x=3$$

Solution Set: $\{2,\ 3\}$

(xiii)

$$(x+2)(x-3)(x+10)(x+5)=-396$$

Pair the factors so the constants add up to the same number: 2 + 5 = 10 + (−3) = 7

$$[(x+2)(x+5)][(x-3)(x+10)]=-396$$
$$[x(x+5)+2(x+5)][x(x+10)-3(x+10)]=-396$$
$$(x^2+5x+2x+10)(x^2+10x-3x-30)=-396$$
$$(x^2+7x+10)(x^2+7x-30)=-396$$

Let

$$x^2+7x=y$$
$$(y+10)(y-30)=-396$$
$$y(y-30)+10(y-30)=-396$$
$$y^2-30y+10y-300+396=0$$
$$y^2-20y+96=0$$
$$y^2-8y-12y+96=0$$
$$y(y-8)-12(y-8)=0$$
$$(y-8)(y-12)=0$$

When y = 8:

$$x^2+7x=8$$
$$x^2+7x-8=0$$
$$x^2-x+8x-8=0$$
$$x(x-1)+8(x-1)=0$$
$$(x-1)(x+8)=0$$
$$x-1=0 \implies x=1$$
$$x+8=0 \implies x=-8$$

When y = 12:

$$x^2+7x=12$$
$$x^2+7x-12=0$$

Here a = 1, b = 7, c = −12:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$
$$=\frac{-7\pm\sqrt{(7)^2-4(1)(-12)}}{2(1)}$$
$$=\frac{-7\pm\sqrt{49+48}}{2}$$
$$x=\frac{-7\pm\sqrt{97}}{2}$$

Solution Set: $\left\{1,\ -8,\ \frac{-7\pm\sqrt{97}}{2}\right\}$

(xiv)

$$x(x-1)(x+2)(x+3)=40$$

Pair the factors: 0 + 2 = 3 + (−1) = 2

$$[x(x+2)][(x-1)(x+3)]=40$$
$$(x^2+2x)[x(x+3)-1(x+3)]=40$$
$$(x^2+2x)(x^2+3x-x-3)=40$$
$$(x^2+2x)(x^2+2x-3)=40$$

Let

$$x^2+2x=y$$
$$y(y-3)=40$$
$$y^2-3y-40=0$$
$$y^2+5y-8y-40=0$$
$$y(y+5)-8(y+5)=0$$
$$(y+5)(y-8)=0$$

When y = −5:

$$x^2+2x=-5$$
$$x^2+2x+5=0$$

Here a = 1, b = 2, c = 5:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$
$$=\frac{-2\pm\sqrt{(2)^2-4(1)(5)}}{2(1)}$$
$$=\frac{-2\pm\sqrt{4-20}}{2}$$

$$= \frac{-2 \pm \sqrt{-16}}{2}$$
$$= \frac{-2 \pm \sqrt{16 \times (-1)}}{2}$$
$$= \frac{-2 \pm \sqrt{16}\ \sqrt{-1}}{2}$$
$$= \frac{-2 \pm 4i}{2}$$
$$= \frac{2(-1 \pm 2i)}{2}$$
$$x = -1 \pm 2i \quad \textit{(where } i = \sqrt{-1}\textit{)}$$

When y = 8:

$$x^2 + 2x = 8$$
$$x^2 + 2x - 8 = 0$$
$$x^2 + 4x - 2x - 8 = 0$$
$$x(x+4) - 2(x+4) = 0$$
$$(x+4)(x-2) = 0$$
$$x + 4 = 0 \implies x = -4$$
$$x - 2 = 0 \implies x = 2$$

Solution Set: $\{2, -4, -1 \pm 2i\}$

(xv)

$$(x-2)(x-6)(x+4)(x+8) + 256 = 0$$

Pair the factors: −2 + 4 = −6 + 8 = 2

$$[(x-2)(x+4)][(x-6)(x+8)] + 256 = 0$$
$$[x(x+4) - 2(x+4)][x(x+8) - 6(x+8)] + 256 = 0$$
$$(x^2 + 4x - 2x - 8)(x^2 + 8x - 6x - 48) + 256 = 0$$
$$(x^2 + 2x - 8)(x^2 + 2x - 48) + 256 = 0$$

Let

$$x^2 + 2x = y$$
$$(y-8)(y-48) + 256 = 0$$
$$y(y-48) - 8(y-48) + 256 = 0$$
$$y^2 - 48y - 8y + 384 + 256 = 0$$
$$y^2 - 56y + 640 = 0$$
$$y^2 - 40y - 16y + 640 = 0$$
$$y(y-40) - 16(y-40) = 0$$
$$(y-16)(y-40) = 0$$

When y = 16:

$$x^2 + 2x = 16$$

$$x^2+2x-16=0$$

Here a = 1, b = 2, c = −16:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$
$$=\frac{-2\pm\sqrt{(2)^2-4(1)(-16)}}{2(1)}$$
$$=\frac{-2\pm\sqrt{4+64}}{2}$$
$$=\frac{-2\pm\sqrt{68}}{2}$$
$$=\frac{-2\pm2\sqrt{17}}{2}$$
$$=\frac{2(-1\pm\sqrt{17})}{2}$$
$$x=-1\pm\sqrt{17}$$

When y = 40:

$$x^2+2x=40$$
$$x^2+2x-40=0$$

Here a = 1, b = 2, c = −40:

$$x=\frac{-2\pm\sqrt{(2)^2-4(1)(-40)}}{2(1)}$$
$$=\frac{-2\pm\sqrt{4+160}}{2}$$
$$=\frac{-2\pm\sqrt{164}}{2}$$
$$=\frac{-2\pm2\sqrt{41}}{2}$$
$$x=-1\pm\sqrt{41}$$

Solution Set: $\{-1\pm\sqrt{17},\ -1\pm\sqrt{41}\}$

Question 3

Solve the following equations by factoring without using substitution.

(i)

$$2x^4-3x^2+1=0$$

Split the middle term: $2 \times 1 = 2 = (-2) \times (-1)$ and $-2-1=-3$

$$2x^4-2x^2-x^2+1=0$$
$$2x^2(x^2-1)-1(x^2-1)=0$$
$$(x^2-1)(2x^2-1)=0$$

Either

$$x^2-1=0$$
$$x^2=1$$
$$\sqrt{x^2}=\sqrt{1}$$
$$x=\pm 1$$

or

$$2x^2-1=0$$
$$2x^2=1$$
$$x^2=\frac{1}{2}$$
$$\sqrt{x^2}=\sqrt{\frac{1}{2}}$$
$$x=\pm\frac{1}{\sqrt{2}}$$

Solution Set: $\left\{\pm 1,\ \pm\frac{1}{\sqrt{2}}\right\}$

(ii)

$$8x^6-7x^3-1=0$$

<u>1st Method (find only real roots)</u>

Split the middle term: $8 \times (-1) = -8 = (-8) \times 1$ and $-8+1=-7$

$$8x^6-8x^3+x^3-1=0$$
$$8x^3(x^3-1)+1(x^3-1)=0$$
$$(x^3-1)(8x^3+1)=0$$

Either

$$x^3-1=0$$
$$x^3=1$$
$$(x^3)^{\frac{1}{3}}=(1)^{\frac{1}{3}}$$

$$x=1$$

or

$$8x^3+1=0$$
$$8x^3=-1$$
$$x^3=-\frac{1}{8}$$
$$(x^3)^{\frac{1}{3}}=(-\frac{1}{8})^{\frac{1}{3}}$$
$$x=(-\frac{1}{2^3})^{\frac{1}{3}}$$
$$x=-\frac{1}{2}$$

Solution Set: $\{1,\ -\frac{1}{2}\}$

2nd Method (find all roots)

$$8x^6-8x^3+x^3-1=0$$
$$8x^3(x^3-1)+1(x^3-1)=0$$
$$(x^3-1)(8x^3+1)=0$$

First factor:

$$x^3-1=0$$
$$x^3-1^3=0$$

Using the formula $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$:

$$(x-1)[(x)^2+(x)(1)+(1)^2]=0$$
$$(x-1)(x^2+x+1)=0$$
$$x-1=0 \implies x=1$$

and $x^2 + x + 1 = 0$, with a = 1, b = 1, c = 1:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$
$$=\frac{-1\pm\sqrt{(1)^2-4(1)(1)}}{2(1)}$$
$$=\frac{-1\pm\sqrt{1-4}}{2}$$
$$=\frac{-1\pm\sqrt{-3}}{2}$$
$$=\frac{-1\pm\sqrt{3\times(-1)}}{2}$$
$$=\frac{-1\pm\sqrt{3}\ \sqrt{-1}}{2}$$

$$x=\frac{-1\pm\sqrt{3}i}{2}$$

Second factor (since $8x^3 = 2^3x^3 = (2x)^3$):

$$8x^3+1=0$$
$$(2x)^3+(1)^3=0$$

Using the formula $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$:

$$(2x+1)[(2x)^2-(2x)(1)+(1)^2]=0$$
$$(2x+1)(4x^2-2x+1)=0$$
$$2x+1=0$$
$$2x=-1$$
$$x=-\frac{1}{2}$$

and $4x^2 - 2x + 1 = 0$, with $a = 4$, $b = -2$, $c = 1$:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$
$$=\frac{-(-2)\pm\sqrt{(-2)^2-4(4)(1)}}{2(4)}$$
$$=\frac{2\pm\sqrt{4-16}}{8}$$
$$=\frac{2\pm\sqrt{-12}}{8}$$
$$=\frac{2\pm\sqrt{12\times(-1)}}{8}$$
$$=\frac{2\pm\sqrt{12}\ \sqrt{-1}}{8}$$
$$=\frac{2\pm2\sqrt{3}i}{8}$$
$$x=\frac{1\pm\sqrt{3}i}{4}$$

Solution Set: $\left\{1,\ -\frac{1}{2},\ \frac{-1\pm\sqrt{3}i}{2},\ \frac{1\pm\sqrt{3}i}{4}\right\}$

(iii)

$$x^2+\frac{1}{x^2}=2$$

Multiply both sides by x^2:

$$x^2\times x^2+\frac{1}{x^2}\times x^2=2x^2$$
$$x^4+1=2x^2$$
$$x^4-2x^2+1=0$$

$$x^4 - x^2 - x^2 + 1 = 0$$
$$x^2(x^2 - 1) - 1(x^2 - 1) = 0$$
$$(x^2 - 1)(x^2 - 1) = 0$$
$$x^2 - 1 = 0$$
$$x^2 = 1$$
$$\sqrt{x^2} = \sqrt{1}$$
$$x = \pm 1$$

Solution Set: $\{\pm 1\}$

(iv)

$$4 \times 2^{2x} - 4 \times 2^x + 1 = 0 \quad \text{...(1)}$$

Let

$$2^x = y$$
$$(2^x)^2 = y^2$$
$$2^{2x} = y^2$$

Put in equation (1):

$$4y^2 - 4y + 1 = 0$$
$$4y^2 - 2y - 2y + 1 = 0$$
$$2y(2y - 1) - 1(2y - 1) = 0$$
$$(2y - 1)(2y - 1) = 0$$
$$2y - 1 = 0$$
$$2y = 1$$
$$y = \frac{1}{2}$$
$$2^x = \frac{1}{2}$$
$$2^x = 2^{-1}$$
$$x = -1$$

Solution Set: $\{-1\}$

Question 4

Write two examples of equations that are not quadratic, but can be written in quadratic form.

Example 1

$$a x^4 + b x^2 + c = 0$$

Let

$$x^2 = y$$
$$(x^2)^2 = y^2$$
$$x^4 = y^2$$

$$a y^2 + by + c = 0 \quad \textbf{(Quadratic form)}$$

Example 2

$$9x^6 - 3x^3 + 7 = 0$$

Let

$$x^3 = y$$
$$(x^3)^2 = y^2$$
$$x^6 = y^2$$

$$9 y^2 - 3 y + 7 = 0 \quad \textbf{(Quadratic form)}$$

Example 3

$$x^4 + 3x^2 - 5 = 0$$

Let

$$x^2 = y$$
$$(x^2)^2 = y^2$$
$$x^4 = y^2$$

$$y^2 + 3y - 5 = 0 \quad \textbf{(Quadratic form)}$$

Question 5

How would you solve the equation $(y - 4)^2 - 7(y - 4) = -6$? Write an explanation and then solve the equation.

Explanation

i. First we reduce the equation to quadratic form.

ii. Take $y - 4 = z$.

iii. Replace $y - 4$ with z.

iv. Solve the quadratic equation: find the value of z (by using the factoring method).

v. Replace z with $y - 4$.

vi. Find the value of y — this is the solution of the original equation.

Solution

$$(y-4)^2-7(y-4)=-6$$

Let

$$y-4=z$$
$$z^2-7z=-6$$
$$z^2-7z+6=0$$
$$z^2-6z-z+6=0$$
$$z(z-6)-1(z-6)=0$$
$$(z-1)(z-6)=0$$

Either

$$z-1=0$$
$$z=1$$
$$y-4=1$$
$$y=1+4$$
$$y=5$$

or

$$z-6=0$$
$$z=6$$
$$y-4=6$$
$$y=6+4$$
$$y=10$$

Solution Set: $\{5,\ 10\}$

Question 6

Solve $y^3 = 125$.

$$y^3 = 125$$
$$y^3 - 125 = 0$$
$$y^3 - 5^3 = 0$$

Using the formula $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$:

$$(y-5)[y^2 + (y)(5) + 5^2] = 0$$
$$(y-5)(y^2 + 5y + 25) = 0$$
$$y - 5 = 0 \implies y = 5$$

and $y^2 + 5y + 25 = 0$, with $a = 1$, $b = 5$, $c = 25$:

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= \frac{-5 \pm \sqrt{(5)^2 - 4(1)(25)}}{2(1)}$$
$$= \frac{-5 \pm \sqrt{25 - 100}}{2}$$
$$= \frac{-5 \pm \sqrt{-75}}{2}$$
$$= \frac{-5 \pm \sqrt{75 \times (-1)}}{2}$$
$$= \frac{-5 \pm \sqrt{25 \times 3}\ \sqrt{-1}}{2}$$
$$y = \frac{-5 \pm 5\sqrt{3}i}{2}$$

Solution Set: $\left\{5,\ \frac{-5+5\sqrt{3}i}{2},\ \frac{-5-5\sqrt{3}i}{2}\right\}$